A. Find Minimum Operations

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: bitmasks, brute force, greedy, math, number theory | https://codeforces.com/problemset/problem/2020/A

You are given two integers n and k.

In one operation, you can subtract any power of k from n. Formally, in one operation, you can replace n by $(n-k^x)$ for any non-negative integer x.

Find the minimum number of operations required to make n equal to 0.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The only line of each test case contains two integers n and k ($1 \le n, k \le 10^9$).

Output

For each test case, output the minimum number of operations on a new line.

Example

Input
6 5 2 3 5 16 4 100 3 6492 10 10 1

Output
2 3 1 4 21 10

Note

In the first test case, $n = 5$ and $k = 2$. We can perform the following sequence of operations:

1. Subtract $2^0 = 1$ from 5. The current value of n becomes 5 - 1 = 4.
2. Subtract $2^2 = 4$ from 4. The current value of n becomes 4 - 4 = 0.

It can be shown that there is no way to make n equal to 0 in less than 2 operations. Thus, 2 is the answer.

In the second test case, $n = 3$ and $k = 5$. We can perform the following sequence of operations:

1. Subtract $5^0 = 1$ from 3. The current value of n becomes 3 - 1 = 2.

2. Subtract $5^0 = 1$ from 2. The current value of *n* becomes 2 - 1 = 1.
3. Subtract $5^0 = 1$ from 1. The current value of *n* becomes 1 - 1 = 0.

It can be shown that there is no way to make *n* equal to 0 in less than 3 operations. Thus, 3 is the answer.